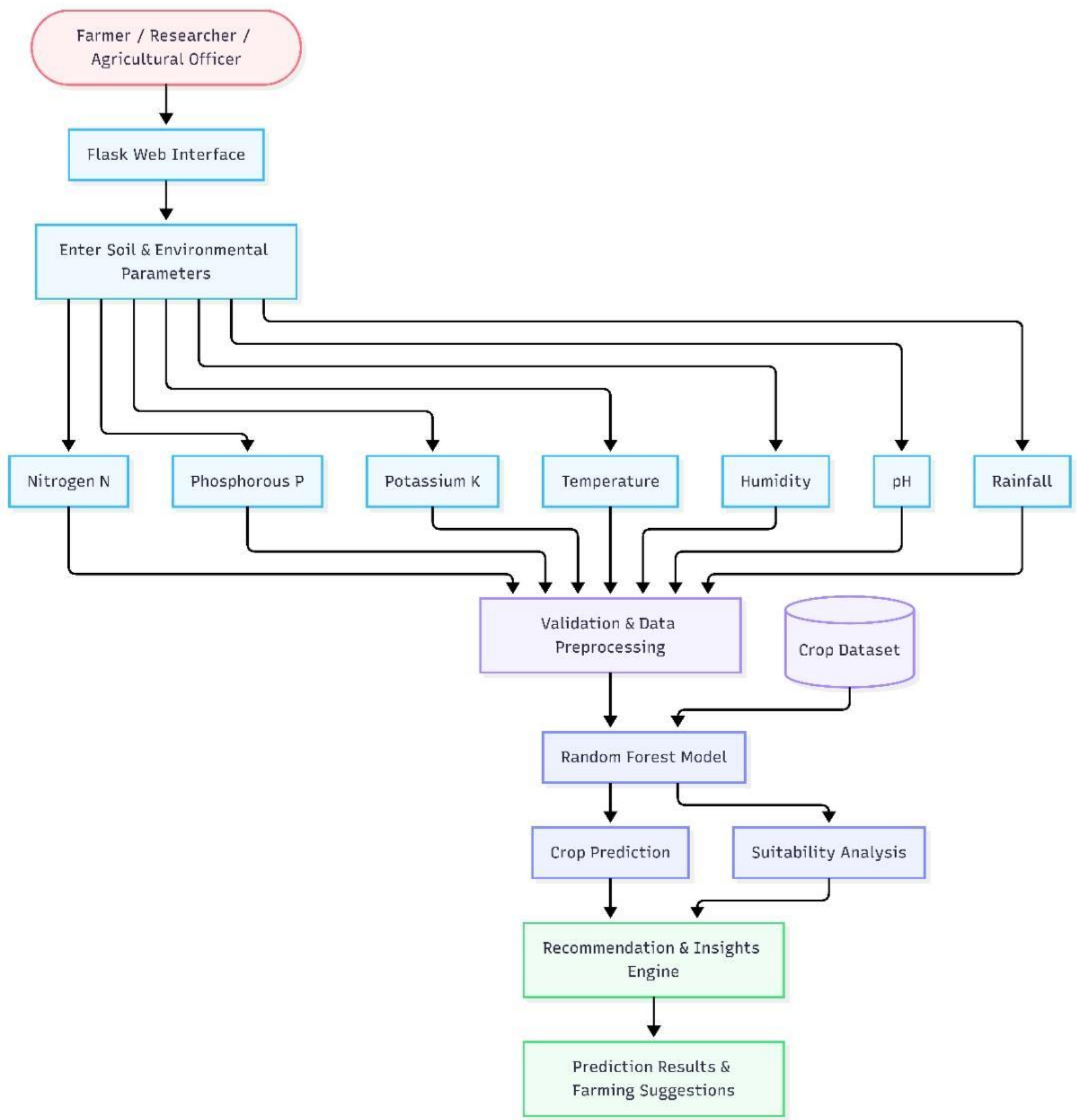


SolutionArchitecture

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram



ComponentDescriptionTable

Component Name	Description /Rolein Architecture	Technologies Used
Presentation Layer	Provides the user interface for farmers to enter crop and soil details and view recommendations.	HTML, CSS, Flask
API Gateway	Processes user requests and connects the frontend with backend services.	Python,Flask
Microservices	Performs crop recommendation fertilizer suggestion, weather analysis, and prediction using ML algorithms.	Python, Scikit-learn, Pandas NumPy
Database	Stores datasets and trained machine learning models used forpredictions.	CSVDataset, Pickle(.pkl)